

DATA VORTEX 2026

Round 1 — Phase 2

Outputs Screenshots

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Note- Refer Notebook to verify outputs (i used duckdb to process sql queries in py notebook)

Link:

https://colab.research.google.com/drive/1FfHbNT_EktquHqC1hfBqJdhPDlrRzvD3?usp=sharing

E3 — Average Engagement by Platform

```
e3 = con.execute("""
SELECT
    platform,
    ROUND(AVG(likes), 2) AS avg_likes,
    ROUND(AVG(shares), 2) AS avg_shares,
    ROUND(AVG(comments), 2) AS avg_comments,
    ROUND(AVG(likes + shares + comments), 2) AS avg_total_engagement
FROM posts
WHERE platform IS NOT NULL
    AND platform <> 'Unknown'
GROUP BY platform
ORDER BY avg_total_engagement DESC;
""").df()
```

e3

	platform	avg_likes	avg_shares	avg_comments	avg_total_engagement
0	YouTube	2548.00	1011.83	504.38	4064.20
1	Instagram	2496.19	1040.84	499.80	4036.83
2	Facebook	2534.89	984.17	506.94	4026.00
3	Reddit	2489.65	1002.23	511.18	4003.05
4	Twitter	2437.86	1005.39	506.13	3949.38

M3 — Most Active Users

```
m3 = con.execute("""
SELECT
    u.user_id,
    u.follower_count,
    u.location,
    COUNT(p.post_id) AS post_count,
    SUM(p.likes + p.shares + p.comments) AS total_engagement
FROM users u
JOIN posts p
    ON u.user_id = p.user_id
GROUP BY
    u.user_id,
    u.follower_count,
    u.location
ORDER BY post_count DESC
LIMIT 10;
""").df()
```

m3

	user_id	follower_count	location	post_count	total_engagement
0	user_zqv2vrf5	13531	Vancouver, Canada	22	90796.0
1	user_nfo3ih5u	40429	Barcelona, Spain	19	77019.0
2	user_n0ok02rt	2531	Dubai, UAE	18	64207.0
3	user_q3vs0uj7	32714	Manchester, UK	18	69039.0
4	user_8wk0j5ae	36943	Los Angeles, USA	18	62937.0
5	user_0irp4abu	44206	Toronto, Canada	17	71943.0
6	user_hdas0iau	1620	Rio de Janeiro, Brazil	16	64006.0
7	user_uerv85na	1824	Rome, Italy	16	70886.0
8	user_8w06w1qu	41923	Melbourne, Australia	16	55783.0
9	user_xetn4exw	14902	Dubai, UAE	16	62175.0

H4 — Follower-to-Engagement Anomaly

Rows returned: 16

	user_id	location	follower_count	total_engagement	engagement_decile
0	user_uerv85na	Rome, Italy	1824	70886.0	1
1	user_n0ok02rt	Dubai, UAE	2531	64207.0	1
2	user_hdas0iau	Rio de Janeiro, Brazil	1620	64006.0	1
3	user_fgjkkrie	Lyon, France	2211	61500.0	1
4	user_r7eg1rac	Houston, USA	2052	56849.0	1
5	user_rr1uzkql	Milan, Italy	1069	55654.0	1
6	user_6wra58f7	Johannesburg, South Africa	4953	55579.0	1
7	user_5oe5i3js	London, UK	3151	55249.0	1
8	user_67hyf45u	Vancouver, Canada	898	52090.0	1
9	user_zwbtkrs	Houston, USA	1009	50627.0	1
10	user_cqailtbuv	Johannesburg, South Africa	3382	50517.0	1
11	user_ogtвуuki	Cairo, Egypt	4459	50212.0	1
12	user_knjtsbf3	Toronto, Canada	1298	48979.0	1
13	user_zozhwbis	Paris, France	4580	48839.0	1
14	user_lfwnie3f	Shanghai, China	2127	48630.0	1
15	user_0aacyvtz	Rome, Italy	1452	48250.0	1